

torch.round#

<https://pytorch.org/docs/stable/generated/torch.round.html>

Description:

Rounds elements of input to the nearest integer. For integer inputs, follows the array-api convention of returning a copy of the input tensor. The return type of output is same as that of input's dtype. This function implements the “round half to even” to break ties when a number is equidistant from two integers (e.g. `round(2.5)` is 2). When the `:attr:`decimals`` argument is specified the algorithm used is similar to NumPy's `around`. This algorithm is fast but inexact and it can easily overflow for low precision dtypes. Eg. `round(tensor([10000], dtype=torch.float16), decimals=3)` is `inf`.

Function Signature

```
torch.round(input, *, decimals=0, out=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> torch.round(torch.tensor((4.7, -2.3, 9.1, -7.7)))
tensor([ 5., -2.,  9., -8.])

>>> # Values equidistant from two integers are rounded towards
the
>>> #   the nearest even value (zero is treated as even)
>>> torch.round(torch.tensor([-0.5, 0.5, 1.5, 2.5]))
tensor([-0.,  0.,  2.,  2.])

>>> # A positive decimals argument rounds to the to that decimal
place
>>> torch.round(torch.tensor([0.1234567]), decimals=3)
tensor([0.1230])

>>> # A negative decimals argument rounds to the left of the
decimal
>>> torch.round(torch.tensor([1200.1234567]), decimals=-3)
tensor([1000.])
```

Notes & Warnings

NOTE

Note This function implements the “round half to even” to break ties when a number is equidistant from two integers (e.g. `round(2.5)` is 2). When the `:attr:`decimals`` argument is specified the algorithm used is similar to NumPy’s `around`. This algorithm is fast but inexact and it can easily overflow for low precision dtypes. Eg. `round(tensor([10000], dtype=torch.float16), decimals=3)` is `inf`.

NOTE

See also `torch.ceil()`, which rounds up. `torch.floor()`, which rounds down. `torch.trunc()`, which rounds towards zero.

Detailed Documentation

Documentation

Rounds elements of input to the nearest integer.

For integer inputs, follows the array-api convention of returning a copy of the input tensor. The return type of output is same as that of input's dtype.

This function implements the “round half to even” to break ties when a number is equidistant from two integers (e.g. `round(2.5)` is 2).

When the `:attr:`decimals`` argument is specified the algorithm used is similar to NumPy's `around`. This algorithm is fast but inexact and it can easily overflow for low precision dtypes. Eg. `round(tensor([10000], dtype=torch.float16), decimals=3)` is `inf`.

`torch.ceil()`, which rounds up. `torch.floor()`, which rounds down. `torch.trunc()`, which rounds towards zero.

`input (Tensor)` – the input tensor.

`decimals (int)` – Number of decimal places to round to (default: 0). If `decimals` is negative, it specifies the number of positions to the left of the decimal point.

`out (Tensor, optional)` – the output tensor.